MH2500 PROBABILITY AND INTRODUCTION TO STATISTICS

NANYANG TECHNOLOGICAL UNIVERSITY

Tutorial 9 Tutorial Time: 28/10/2021

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We will discuss several questions from Chapter 7 in Ross' book.

Tutorial Questions

(1) Consider n independent flips of a coin having probability p of landing on heads. Say that a changeover occurs whenever an outcome differs from the one preceding it. For instance, if $n = 5$ and the outcome is $HHTHT$, then there are 3 changeovers. Find the expected number of changeovers.

(2) The joint density of X and Y is given by

$$f(x, y) = \frac{e^{-y}}{y}, \quad 0 < x < y, 0 < y < \infty$$

Compute $\mathbb{E}[X^3|Y = y]$.

(3) A certain region is inhabited by r distinct types of a certain species of insect. Each insect caught will, independently of the types of the previous catches, be of type i with probability

$$P_i, \ i = 1, \dots, r, \quad \sum_{i=1}^{r} P_i = 1.$$

(a) Compute the mean number of insects that are caught before the first type 1 catch.
(b) Compute the mean number of types of insects that are caught before the first type 1 catch.

(4) The random variables X and Y have a joint density function given by

$$f(x, y) = \begin{cases} 2e^{-2x}/x, & 0 \le x < \infty, 0 \le y \le x \\ 0, & \text{otherwise} \end{cases}$$

Compute $Cov(X, Y)$.

(5) The joint density function of X and Y is given by

$$f(x, y) = \frac{1}{y} e^{-(y+x/y)}, \quad x > 0, y > 0$$

Find $\mathbb{E}(X)$, $\mathbb{E}(Y)$, and show that $Cov(X, Y) = 1$.